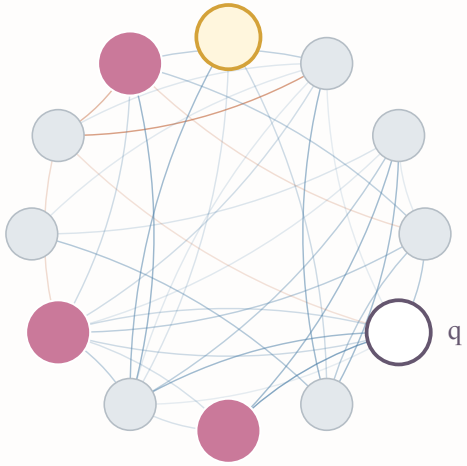
Predictive risk and recording decisions

in partially observed neural networks

Nominal versus prior-averaged local risk for stimulus-response prediction



12 simulated units · M=3

Known connectivity; uncertain physiology

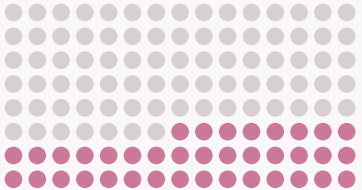
Blue: positive; orange: negative connections

- Recorded node
- Stimulated node
- Response readout (one shown)

1 Recording agreement

Nominal vs prior-averaged local risk
40 independent networks

68% 82 of 120 single-target designs are identical



65% 26 of 40 shared-target designs are identical



same recording set

2 Response error

Common MAP fit and budget
Same 40 networks

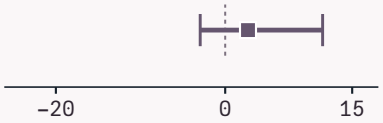
Single response

-3.7% [-18.5, +12.1]



Shared three-response

+2.7% [-3.0, +11.5]



Mean-MSE change vs nominal (%)

97.5% paired network-cluster CI

3 Risk forecasts

Fixed data and point prediction
40 new independent networks

19 of 40 favour prior averaging; 21 favour nominal

$$S(v,e)=\log v+e^2/v$$

v: MSE forecast; e: response error

One network contributes 64% of the summed score difference

MSE-forecast score difference

-0.46 [-1.16, +0.02]



Prior-averaged minus nominal

95% paired network-cluster CI

Integrated stimulus response

$$q=\int \Delta r(t)\,dt$$

$$\Delta r(t)=r_{\text{stim}}(t)-r_{\text{control}}(t)$$

Identical recording sets and data yield identical predictions under the common MAP fit.
All primary comparison intervals include zero.

Exploratory 100-unit rank-two family (8 independent networks): 4/8 setting-specific 95% intervals exclude zero, all favouring prior averaging. Intervals are unadjusted; one point estimate favours nominal.